# 8.4 Proportionality Theorems

Name _____ Period ____ Date _____

Geometry 8.4 (30-minute plan) · Student sheet

**I can…**

- use a segment parallel to one side of a triangle to find a missing length.
- decide whether a segment is parallel to a side of a triangle.
- use parallel lines and angle bisectors to find missing lengths.

**Vocabulary** · fill in the missing word

**Triangle Proportionality Theorem** — a line _____ to one side of a triangle that crosses the other two sides divides them proportionally

**Converse** — if a line divides two sides of a triangle _____, it is parallel to the third side

**Transversal** — a line that _____ two or more lines

**Angle bisector** — a ray that divides an angle into two _____ angles

## Part 1 · Notes: fill in the blanks

**Triangle Proportionality Theorem:** if a line _____ to one side of a triangle intersects the other two sides, then it divides the two sides _____.

**Converse:** if a line divides two sides of a triangle proportionally, then it is _____ to the third side.

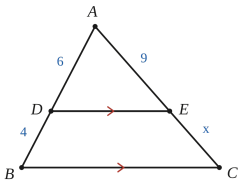
**Three Parallel Lines Theorem:** if three parallel lines intersect two _____, then they divide the transversals proportionally.

**Triangle Angle Bisector Theorem:** if a ray bisects an angle of a triangle, then it divides the opposite side into segments proportional to the other two _____.

## Part 2 · Examples (work them with the class)

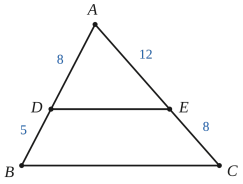
### Example 1

In △ABC, $\overline{DE} \parallel \overline{BC}$. AD = 6, DB = 4, and AE = 9. Find EC.



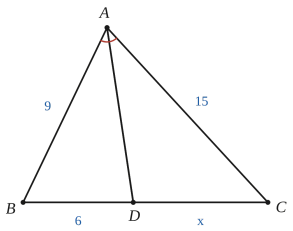

### Example 2

In △ABC, AD = 8, DB = 5, AE = 12, and EC = 8. Is $\overline{DE} \parallel \overline{BC}$?



### Example 3

$\overline{AD}$ bisects $\angle BAC$. AB = 9, AC = 15, and BD = 6. Find DC.




## Part 3 · Partner practice

### Partner practice

In $\triangle ABC$, $\overline{DE} \parallel \overline{BC}$. AD = 8, DB = 12, and EC = 15. Find AE.

